Status	Finished
Started	Monday, 26 October 2026, 6:46 PM
Completed	Monday, 26 October 2026, 6:56 PM
Duration	9 mins 39 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 1.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Merge Sort
- b. Quick Sort
- c. Linear Search
- d. Bubble Sort

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Rearranging data without sorting
- b. Sorting data in a single pass
- c. Dividing the input into parts, solving each part, and combining the solutions
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **3**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Makes code execution slower
- b. Decreases the efficiency of algorithms
- c. Makes data harder to manage
- d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **4**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

- a. Dividing the input into smaller subproblems
- b. Comparing each element with the others
- c. Combining sorted sublists
- d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

- a. It allows for efficient parallel processing and sorting of data
- b. It avoids the need for recursion
- c. It only works on small datasets
- d. It simplifies the sorting process by using only one pass

The correct answer is: It allows for efficient parallel processing and sorting of data

Question **6**

Complete

Mark 0.00 out of 1.00

Very slow way of sorting is_____

- a. Heap sort
- b. Quick sort
- c. Bubble sort
- d. Insertion sort

The correct answer is:
Insertion sort

Question **7**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

- a. Sorting
- b. Rearranging
- c. Searching
- d. Merging

The correct answer is: Sorting

Question **8**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Dynamic programming
- b. Divide and conquer
- c. Backtracking
- d. Greedy method

The correct answer is:
Divide and conquer

Question **9**

Complete

Mark 1.00 out of 1.00

What is mean by stable sorting algorithm?

- a. A sorting algorithm is stable if it preserves the order of all keys
- b. A sorting algorithm is stable if it preserves the order of duplicate keys
- c. A sorting algorithm is stable if it doesn't preserver the order of duplicate keys
- d. A sorting algorithm is stable if it preserves the order of non-duplicate keys

The correct answer is:
A sorting algorithm is stable if it preserves the order of duplicate keys

Question **10**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

- a. All of the above
- b. Searching
- c. Duplicates
- d. Selection

The correct answer is: All of the above

Question **11**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

- a. Binary Search
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **12**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is described as making multiple passes through a list, comparing elements, and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Bubble Sort
- c. Insertion Sort
- d. Quick Sort

The correct answer is: Bubble Sort

Question **13**

Complete

Mark 1.00 out of 1.00

Which Python function would you use to sort a list in-place?

- a. arrange()
- b. sorted()
- c. sort()
- d. order()

The correct answer is: sort()

Question **14**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It complicates the analysis process
- c. It has no effect on the analysis process
- d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **15**

Complete

Mark 1.00 out of 1.00

What is one of the key advantages of using the built-in sorted() function in Python?

- a. It only works with integer arrays
- b. It sorts data out of the box efficiently
- c. It is less efficient than custom sorting algorithms
- d. It requires external libraries

The correct answer is: It sorts data out of the box efficiently